

Event Handling Overview

Definition: The process of handling events is known as event handling.

Java uses “delegation event model” to process the events raised in a GUI program

Delegation Event Model

The *delegation event model* provides a consistent way of generating and processing events. In this model a *source* generates an event and sends it to one or more *listeners*. A listener waits until it receives an event and once it receives an event, processes the event and returns.

The advantage of this model is the application logic for processing the events is clearly separated from the user interface logic. In this model the listeners must register with a source to receive the events.

Another advantage of this model is, notification of events is sent only to the registered listeners.

Events

An *event* is an object that describes a state change in the source. Some examples of activities that cause events to be generated are clicking a button, pressing a key on the keyboard, selecting an item in the list.

Event Sources

A *source* is an object that generates an event. A source might generate more than one event. Examples of sources are button, check box, radio button, text box etc.

A source must be registered with listeners in order for the events to be detected. The general form of a registration method is as follows:

```
public void addTypeListener(TypeListener tl)
```

A listener can also be unregistered from the source using the remove method whose syntax is as follows:

```
public void removeTypeListener(TypeListener tl)
```

Event Listeners

A *listener* is an object that is notified when an event occurs. A listener has two major requirements. First is, listener must be registered with a source to receive notifications and second is, it must implement methods to receive and process these notifications.

In Java, events and sources are maintained as classes, listeners are maintained as interfaces. Most of them are available in *java.awt.event* package.

Examples of listeners are *MouseListener*, *Key Listener* etc.

java.awt.event Package

The *java.awt.event* package contains many event classes which can be used for event handling.

Root class for all the event classes in Java is **EventObject** which is available in *java.util* package.

Root class for all AWT event classes is **AWTEvent** which is available in *java.awt* package and is a sub class of *EventObject* class.

Some of the frequently used event classes available in *java.awt.event* package are listed below:

Event Class	Description
ActionEvent	Generated when a button is pressed, a list item is double-clicked, or a menu item is selected.
AdjustmentEvent	Generated when a scroll bar is manipulated.
ComponentEvent	Generated when a component is hidden, moved, resized, or becomes visible.
ContainerEvent	Generated when a component is added to or removed from a container.
FocusEvent	Generated when a component gains or loses keyboard focus.
InputEvent	Abstract superclass for all component input event classes.
ItemEvent	Generated when a check box or list item is clicked; also occurs when a choice selection is made or a checkable menu item is selected or deselected.
KeyEvent	Generated when input is received from the keyboard.
MouseEvent	Generated when the mouse is dragged, moved, clicked, pressed, or released; also generated when the mouse enters or exits a component.
MouseWheelEvent	Generated when the mouse wheel is moved.
TextEvent	Generated when the value of a text area or text field is changed.
WindowEvent	Generated when a window is activated, closed, deactivated, deiconified, iconified, opened, or quit.

Adapter Classes

Adapter classes are a set of classes which can be used to simplify the event handling process. As we can see in the mouse event handling program [here](#), even though we want to handle only one or two mouse events, we have to define all other remaining methods available in the listener interface(s).

To remove the aforementioned disadvantage, we can use adapter classes and define only the required event handling method(s). Some of the adapter classes available in the *java.awt.event* package are listed below:

Adapter Class	Listener Interface
ComponentAdapter	ComponentListener
ContainerAdapter	ContainerListener
FocusAdapter	FocusListener
KeyAdapter	KeyListener
MouseAdapter	MouseListener and (as of JDK 6) MouseMotionListener and MouseWheelListener
MouseMotionAdapter	MouseMotionListener
WindowAdapter	WindowListener, WindowFocusListener, and WindowStateListener